What momentum says about impact

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Learning target: Connect stopping impulse to average force without claiming a complete damage forecast.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 9.*
They exchange velocities: A=-3 m/s; B=+5 m/s.

02. Worked example: explain the model
Momentum arrows equal; lighter object's velocity twice as long.

03. Read and interpret the model
The .1 s stop has 10 times the average force.

04. Apply the physics
Initial momentum = 40 kg·m/s. Average stopping-force magnitude = 40/4 = 10 N, opposite the motion.

05. Explain your reasoning
Need energy deposition, trajectory, atmosphere, contact/deformation area, material response and geography; simplistic stopping time may be unjustified.

06. Review: connect an earlier idea *Review: Unit 2, Day 16.*
g=4 N/kg; force 32 N inward.

07. Exit check
Both double.

08. STRETCH: test the model
Each magnitude 24 N average; contact histories and material response may differ.

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01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 9.*
v_f=1 m/s; K_i=51 J, K_f=3 J. K decreases by 48 J; perfectly inelastic.

02. Worked example: explain the model
Momentum arrows equal; lighter object's velocity twice as long.

03. Read and interpret the model
The .1 s stop has 10 times the average force.

04. Apply the physics
Impulse 2×10^8 N·s; force 4×10^8 N; 5 s stop gives 4×10^7 N.

05. Explain your reasoning
Need energy deposition, trajectory, atmosphere, contact/deformation area, material response and geography; simplistic stopping time may be unjustified.

06. Review: connect an earlier idea *Review: Unit 2, Day 16.*
v=6 m/s; a≈2.25 m/s²; T≈16.76 s. Speed constant, direction changing.

07. Exit check
A speed half; $KA=p^2/(2\ mA)=KB/2$.

08. STRETCH: test the model
Stop: -5 N·s, -125 N. Rebound: -8 N·s, -200 N; direction reversal increases momentum change.